

Ocean Waves: Time-Domain Analysis of Drag Forces on a Submerged Spherical Housing

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Semester Task | Module: Ocean Waves | **Lecturers:** Prof. Dr.-Ing. Sascha Kosleck, Dr.-Ing. Christoph Otto | **Date:** July 2026

Abstract

This study investigates the horizontal wave-induced drag force acting on a fixed spherical camera housing mounted beneath a stationary offshore platform. The prescribed fully developed wind-generated sea, characterized by a significant wave height of 0.98 m and a peak period of 10 s, was represented using a Bretschneider spectrum. The spectrum was discretized into 100 harmonic components with a frequency increment of 0.005 Hz and a maximum frequency of 0.50 Hz. Integration of the discrete spectrum produced a significant wave height of 0.979 m, corresponding to a relative error of approximately 0.098%, thereby confirming adequate preservation of the target wave energy. Based on the nondimensional wave steepness and relative water depth, linear Airy wave theory was selected to determine the horizontal fluid velocity at the housing center, located 2 m below the mean water level. Three phase configurations were examined over the 200 s repetition period of the discretized wave field. The zero-phase configuration produced the largest absolute velocity and drag force, reaching $1.622 \text{ m}\cdot\text{s}^{-1}$ and 118.61 N, respectively. The random and period-based phase configurations resulted in substantially lower maximum forces of 13.18 N and 12.35 N. These results demonstrate that phase coherence strongly influences instantaneous extreme loads and that the zero-phase configuration provides the governing, although highly conservative, design condition.

1. Introduction

Marine observation systems are commonly installed on offshore structures to monitor the surrounding environment. Such equipment is continuously exposed to ocean waves, which generate time-dependent fluid velocities and corresponding hydrodynamic loads. Reliable estimation of these loads is therefore necessary for the structural design of the supporting components. The mechanics of surface waves and the associated subsurface particle kinematics provide the theoretical basis for predicting these effects. Depending on the wave height, period, and local water depth, an appropriate wave theory must be selected to describe the wave-induced velocity field. The resulting kinematics can subsequently be used to estimate the forces acting on fixed submerged bodies.

The present study investigates the horizontal wave-induced drag force acting on a spherical camera housing mounted below a stationary maritime platform, as illustrated in Figure 1. The housing has a diameter of $d_h = 0.50$ m, and its centre is located at $z_h = -2.0$ m relative to the mean water level. It is connected to the platform by a supporting rod whose required load capacity is to be assessed. The local water depth is $h = 98.1$ m, while the platform is assumed to remain at rest. Consequently, motions of the platform and their influence on the relative fluid velocity are neglected.

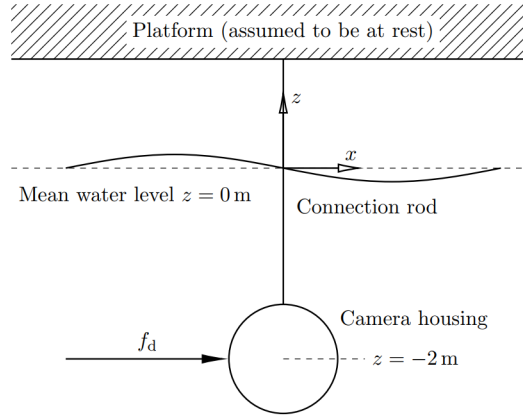


Figure 1: Spherical camera housing mounted below a stationary platform in the ocean.

The environmental conditions correspond to a fully developed wind-generated sea with no swell and no limitation of the wind fetch. The significant wave height and peak period are given as $H_s = 0.98$ m and $T_p = 10$ s, respectively. Under these conditions, the irregular sea state is represented by a Bretschneider spectrum, which is defined directly in terms of H_s and T_p . The principal parameters used in the analysis are summarized in Table 1.

The Bretschneider spectrum is discretized into harmonic wave components over the frequency range from 0.005 to 0.50 Hz. The accuracy of this discretisation is examined by integrating the discrete energy-density spectrum and reconstructing the corresponding significant wave height. Based on the nondimensional wave steepness and relative water depth, linear Airy wave theory is applied to calculate the horizontal fluid velocity at the location

of the camera housing. The applicability of this theory is evaluated using the wave-theory classification proposed by Le Méhauté [?].

Table 1: Principal environmental and structural parameters

Quantity	Symbol	Value	Unit
Significant wave height	H_s	0.98	m
Peak period	T_p	10	s
Local water depth	h	98.1	m
Housing-center elevation	z_h	-2.0	m
Housing diameter	d_h	0.50	m
Seawater density	ρ	1020	$\text{kg} \cdot \text{m}^{-3}$
Drag coefficient	C_D	0.45	–
Gravitational acceleration	g	9.81	$\text{m} \cdot \text{s}^{-2}$
Maximum spectral frequency	$f_{\max}$	0.50	Hz
Frequency increment	Δf	0.005	Hz

Because an irregular wave field is formed by superimposing harmonic components, the selected phase shifts strongly influence the instantaneous surface elevation, fluid velocity, and resulting force. Three prescribed phase configurations are therefore considered: zero phase shifts, uniformly distributed random phase shifts, and period-based phase shifts defined by $\phi_i = 2\pi T_i^2$. For each configuration, the surface elevation at $x = 0$, the horizontal velocity at the housing location, and the horizontal drag force are evaluated over one complete repetition period of the discretised wave field.

Only the horizontal drag contribution is included in the present analysis. The drag force is calculated from the instantaneous horizontal velocity using the projected area of the spherical housing. Wave-induced inertia forces, ambient currents, structural motion, and interactions with the supporting rod are outside the scope of the simplified model. The primary objective is to quantify the effect of the prescribed phase configurations on the maximum horizontal velocity and drag force and to identify a suitable design load for the connection rod.

2. Solution and Results

2.1. Wave Spectrum and Discretization

The specified environmental conditions describe a fully developed wind-generated sea without fetch limitation or swell. Since both the significant wave height H_s and peak period T_p are prescribed, the Bretschneider spectrum was selected to represent the sea state [?]. Its angular-frequency form is

$$S(\omega) = \frac{5}{16} H_s^2 \omega_p^4 \omega^{-5} \exp \left[-\frac{5}{4} \left(\frac{\omega_p}{\omega} \right)^4 \right] \quad (1)$$

where $\omega_p = 2\pi/T_p$ is the peak angular frequency. The spectrum was discretised from $f_{\min} = 0.005$ Hz to $f_{\max} = 0.50$ Hz, using a constant frequency increment of $\Delta f = 0.005$ Hz. This procedure produced 100 harmonic wave components. The amplitude assigned to each component was calculated as

$$\hat{a}_i = \sqrt{2S(\omega_i)\Delta\omega}, \quad \Delta\omega = 2\pi\Delta f \quad (2)$$

Figure 2 shows the resulting energy-density spectrum and the corresponding discrete wave amplitudes. Both distributions attain their maximum near $\omega_p = 0.628 \text{ rad} \cdot \text{s}^{-1}$, corresponding to the prescribed peak period of 10 s. Unlike $S(\omega)$, which represents energy density per unit angular frequency, $\hat{a}_i$ represents the physical amplitude of an individual discrete wave component and therefore depends on the selected frequency increment.

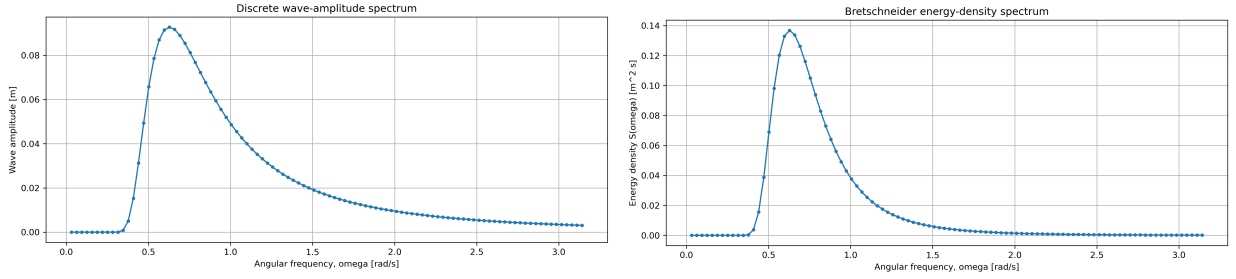


Figure 2: Bretschneider energy-density spectrum and corresponding discrete wave-component amplitudes.

2.2. Verification of the Discretized Spectrum

The zeroth spectral moment was evaluated using rectangular integration,

$$m_0 = \sum_{i=1}^n S(\omega_i)\Delta\omega \quad (3)$$

and the corresponding significant wave height was reconstructed from

$$H_{s,\text{res}} = 4\sqrt{m_0} \quad (4)$$

The numerical integration yielded $m_0 = 0.05991 \text{ m}^2$ and $H_{s,\text{res}} = 0.97904 \text{ m}$. Compared with the prescribed value of 0.98 m, the relative deviation is only 0.098%. The close agreement confirms that the selected frequency range and discretisation accurately preserve the energy of the target sea state.

2.3. Selection of Wave Theory

The applicability of linear wave theory was assessed using the nondimensional wave steepness and relative water depth,

$$\frac{H_s}{gT_p^2} = 9.99 \times 10^{-4}, \quad \frac{h}{gT_p^2} = 0.100 \quad (5)$$

When plotted on the Le Méhauté applicability diagram, these parameters lie within the linear-wave theory region [?]. Linear Airy theory was therefore adopted to calculate the wave-induced kinematics [?]. The wave number of every component was obtained from the dispersion relation

$$\omega_i^2 = gk_i \tanh(k_i h) \quad (6)$$

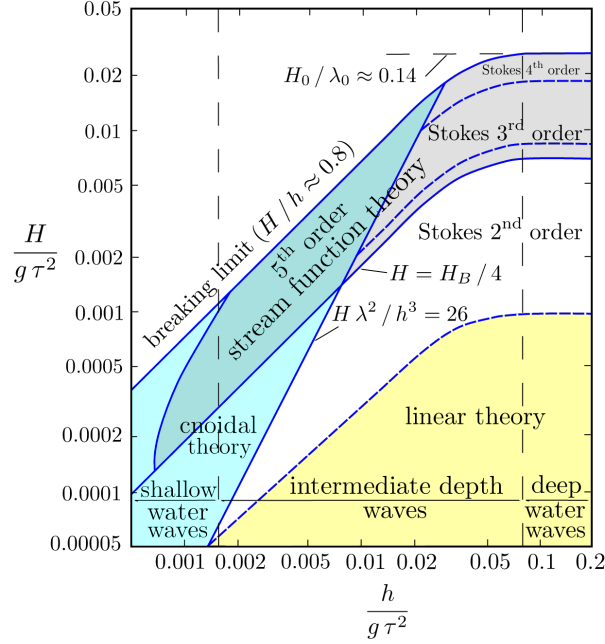


Figure 3: Position of the investigated sea state in the Le Méhauté wave-theory applicability diagram (adapted from Le Méhauté% [?], licensed under CC BY 4.0 / CC BY-SA).

For a superposition of n harmonic components, the surface elevation at $x = 0$ and the horizontal velocity at the housing position $z = -2$ m were evaluated from

$$\eta(x, t) = \sum_{i=1}^n \hat{a}_i \cos(\theta_i) \quad (7)$$

$$u_x(x, z, t) = \sum_{i=1}^n \hat{a}_i \omega_i \frac{\cosh[k_i(z + h)]}{\sinh(k_i h)} \cos(\theta_i) \quad (8)$$

where $\theta_i = k_i x - \omega_i t + \phi_i$. The horizontal drag force was subsequently calculated as

$$F_{D,x}(t) = \frac{1}{2} \rho C_D A_{\text{ref}} u_x(t) |u_x(t)|, \quad A_{\text{ref}} = \frac{\pi d_h^2}{4} \quad (9)$$

2.4. Phase Configurations and Time-Domain Response

Three prescribed phase configurations were considered:

- **Case A:** $\phi_i = 0$
- **Case B:** $\phi_i = 2\pi \cdot \text{rnd}([0, 1))$
- **Case C:** $\phi_i = 2\pi T_i^2$

Since all frequencies are integer multiples of Δf , the reconstructed wave field repeats after

$$T_{\max} = \frac{1}{\Delta f} = 200 \text{ s} \quad (10)$$

The analysis was therefore conducted from 0 to 200 s. A time increment of $\Delta t = 0.2$ s provided ten time steps per cycle at the maximum frequency, resulting in 1001 time samples.

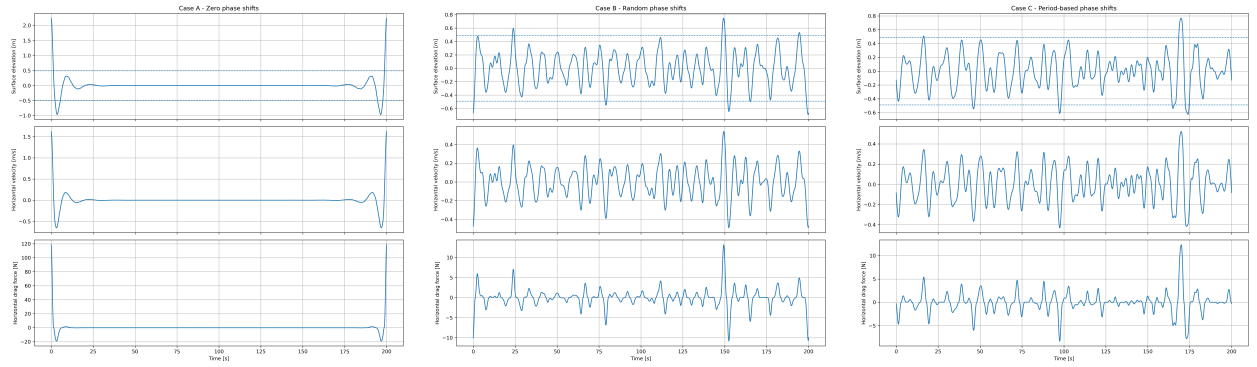


Figure 4: Time histories of surface elevation, horizontal fluid velocity and horizontal drag force for (a) zero phase shifts, (b) random phase shifts, and (c) period-based phase shifts.

Case A produces a strongly focused response at the beginning and end of the analysis interval because all components are initially in phase. The response at $t = 200$ s repeats that at $t = 0$, as expected from the periodicity of the discretised spectrum. The maximum absolute surface elevation, horizontal velocity and drag force are 2.244 m, $1.622 \text{ m} \cdot \text{s}^{-1}$ and 118.61 N, respectively. This configuration represents an artificial but highly conservative condition of complete phase coherence.

Cases B and C exhibit irregular time histories because their components are distributed over different phases. For Case B, the maximum absolute surface elevation is 0.748 m, while the maximum velocity and force are $0.541 \text{ m} \cdot \text{s}^{-1}$ and 13.18 N. Case C gives similar values of 0.767 m, $0.524 \text{ m} \cdot \text{s}^{-1}$ and 12.35 N. The dashed lines shown at $\eta = \pm H_s/2$ are reference levels only and should not be interpreted as physical bounds on the instantaneous surface elevation.

2.5. Comparison of the Three Phase Configurations

The comparison demonstrates that phase selection has a decisive influence on the extreme instantaneous response, although the underlying energy-density spectrum remains unchanged. Cases B and C produce distributed irregular oscillations of comparable magnitude, whereas Case A concentrates the contributions of all components into a short interval. The maximum values are summarized in Table 2.

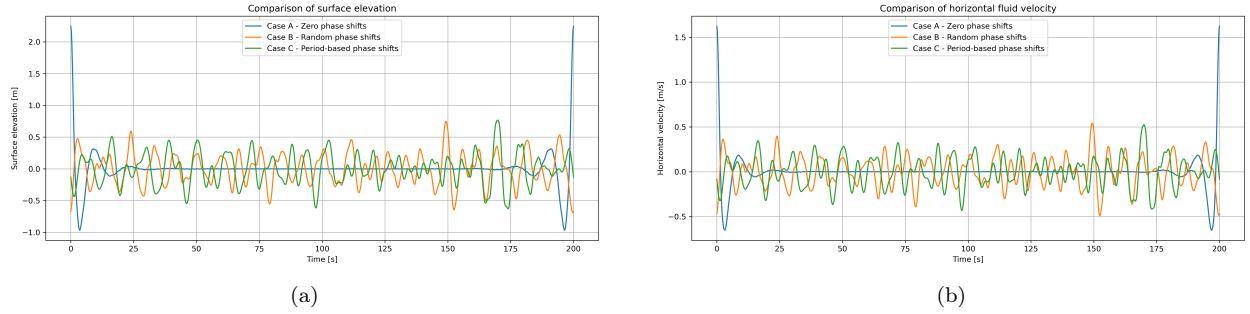


Figure 5: Comparison of (a) surface elevation and (b) horizontal fluid velocity for the three phase configurations.

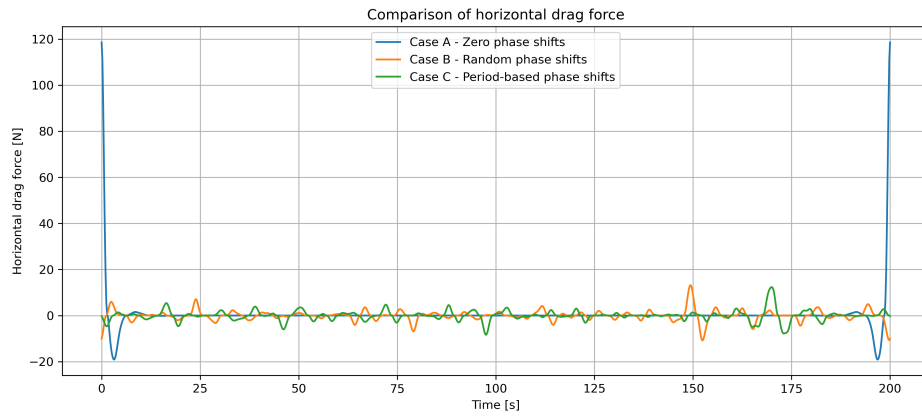


Figure 6: Comparison of horizontal drag forces for the three phase configurations.

Table 2: Maximum absolute responses for the prescribed phase configurations.

Phase configuration	Maximum surface elevation (m)	Maximum horizontal velocity ($\text{m} \cdot \text{s}^{-1}$)	Maximum horizontal drag force (N)
Case A – zero phases	2.244	1.622	118.61
Case B – random phases	0.748	0.541	13.18
Case C – period-based phases	0.767	0.524	12.35

The maximum velocity in Case A is approximately three times that obtained for the random-phase case. Since the drag force is proportional to $|u_x|u_x$, this velocity amplification increases the maximum force by approximately a factor of nine. Cases B and C differ only slightly, indicating that both phase distributions prevent the systematic constructive interference observed in Case A. Among the prescribed configurations, the zero-phase condition consequently governs the maximum calculated connection-rod load.

3. Conclusion

The horizontal wave-induced loading on the submerged spherical camera housing was evaluated by combining a discretized Bretschneider spectrum with linear Airy wave theory and a quadratic drag force formulation. The reconstructed significant wave height of 0.979 m differed from the prescribed value of 0.98 m by only approximately 0.098%, confirming that the selected frequency range and discretization accurately represented the target sea state. Furthermore, the nondimensional wave parameters placed the investigated condition within the applicability range of linear wave theory.

Although the three simulations were based on the same energy-density spectrum, their instantaneous responses differed substantially because of the selected phase shifts. Complete phase alignment in Case A caused strong constructive interference at the beginning and end of the 200 s repetition interval. This produced a maximum absolute surface elevation of 2.244 m, a horizontal velocity of $1.622 \text{ m} \cdot \text{s}^{-1}$, and a drag force of 118.61 N. In comparison, the random-phase and period-based configurations produced irregular but distributed responses, with maximum drag forces of 13.18 N and 12.35 N, respectively.

The results confirm that the phase configuration is a decisive factor in estimating extreme instantaneous wave loads. Because the drag force varies with the square of the horizontal velocity, the velocity amplification generated by phase focusing leads to a disproportionately large increase in force. Consequently, the zero-phase condition governs the calculated connection-rod load and may be adopted as a conservative design case. However, it represents an artificial condition of complete phase coherence rather than a typical irregular sea realization. For a probabilistic design assessment, several independent random-phase realizations, together with inertia effects, currents, structural motion, and loading on the supporting rod, should also be considered.